KAIZEN MATHS

Lesson starter

Taylor and Maclaurin Series

Taylor Series

Name

S1

Use Taylor's formula about the given value of a.

Given f(-1) = 2, f'(-1) = 6, f''(-1) = 6 and f'''(-1) = -24, find the Taylor polynomial about x = -1 up to $(x - -1)^3$.

S2

Find each Taylor expansion about the stated point.

Find the Taylor expansion of 1 x about x = 2 up to the cubic term.

S3

Use a Taylor expansion about the stated point to approximate the value.

Use a Taylor expansion to approximate $e^{1.1}$.

S4

Compare coefficients with Taylor's formula to find derivative values.

A Taylor expansion about x = -1 is $f(x) = 0 + 5(x - -1) + 4(x - -1)^2 - 1(x - -1)^3 + \ldots$. Find f(-1), f'(-1), f''(-1) and f'''(-1).

Show your working

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KAIZEN MATHS

Exit ticket

Taylor and Maclaurin Series

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Name

EA1

Use Taylor's formula about the given value of a.

Given f(1) = 0, f'(1) = 5, f''(1) = -8 and f'''(1) = 12, find the Taylor polynomial about x = 1 up to $(x - 1)^3$.

EA2

Find each Taylor expansion about the stated point.

Find the Taylor expansion of ln x about x = 1 up to the cubic term.

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Exit ticket

Taylor and Maclaurin Series

Taylor Series

Name

EB1

Use a Taylor expansion about the stated point to approximate the value.

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Teacher answers & working

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S1

Use Taylor's formula about the given value of a.

Given f(-1) = 2, f'(-1) = 6, f''(-1) = 6 and f'''(-1) = -24, find the Taylor polynomial about x = -1 up to $(x - -1)^3$.

Answer:

$2 + 6(x - -1) + 3(x - -1)^2 - 4(x - -1)^3$

Working:

Use $f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$.

Here a = -1.

Substitute the derivative values: $f(x) = 2 + 6(x - -1) + \frac{6(x--1)^2}{2!} + \frac{-24(x--1)^3}{3!}$.

Simplify each coefficient.

$f(x) = 2 + 6(x - -1) + 3(x - -1)^2 - 4(x - -1)^3$.

S2

Find each Taylor expansion about the stated point.

Find the Taylor expansion of 1 x about x = 2 up to the cubic term.

Answer:

$\frac{1}{2} - \frac{(x-2)}{4} + \frac{(x-2)^2}{8} - \frac{(x-2)^3}{16}$

Working:

Write x = 2 + h, so h = x - 2.

Then $\frac{1}{x} = \frac{1}{2+h} = \frac{1}{2}\frac{1}{1+\frac{h}{2}}$.

Use $\frac{1}{1+u} = 1 - u + u^2 - u^3 + \ldots$ with $u = \frac{h}{2}$.

So $\frac{1}{1+\frac{h}{2}} = 1 - \frac{h}{2} + \frac{h^2}{4} - \frac{h^3}{8} + \ldots$.

Multiply by $\frac{1}{2}$ and then replace h by x - 2.

Answer: $\frac{1}{2} - \frac{(x-2)}{4} + \frac{(x-2)^2}{8} - \frac{(x-2)^3}{16}$.

Teacher answers & working

Taylor and Maclaurin Series · Taylor Series

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S3

Use a Taylor expansion about the stated point to approximate the value.

Use a Taylor expansion to approximate $e^{1.1}$.

Answer:

3.004154

Working:

Use the Taylor expansion for e^x about x = 1.

Let h be the distance from the expansion point. Here h = 0.1.

Use $e(1 + h + \frac{h^2}{2!} + \frac{h^3}{3!})$.

Substitute h = 0.1.

$e^{1.1} \approx e(1 + 0.1 + \frac{(0.1)^2}{2!} + \frac{(0.1)^3}{3!})$.

Calculate this expression to get $e^{1.1} \approx 3.004154$.

S4

Compare coefficients with Taylor's formula to find derivative values.

A Taylor expansion about x = -1 is f(x) = 0 + 5(x - -1) + 4(x - -1)^2 - 1(x - -1)^3 + … . Find f(-1), f'(-1), f''(-1) and f'''(-1).

Answer:

0, 5, 8, -6

Working:

Compare with f(x) = f(a) + f'(a)(x-a) + $\frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$ + … .

The constant term is f(-1) = 0.

The coefficient of (x - -1) is f'(-1) = 5.

The coefficient of (x - -1)^2 is $\frac{f''(-1)}{2!}$, so f''(-1) = 2 × 4 = 8.

The coefficient of (x - -1)^3 is $\frac{f'''(-1)}{3!}$, so f'''(-1) = 6 × -1 = -6.

EA1

Use Taylor's formula about the given value of a.

Given f(1) = 0, f'(1) = 5, f''(1) = -8 and f'''(1) = 12, find the Taylor polynomial about x = 1 up to (x - 1)^3.

Answer:

5(x - 1) - 4(x - 1)^2 + 2(x - 1)^3

Working:

Use $f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!}$.

Here a = 1.

Substitute the derivative values: f(x) = 0 + 5(x - 1) + $\frac{-8(x-1)^2}{2!} + \frac{12(x-1)^3}{3!}$.

Simplify each coefficient.

$f(x) = 5(x - 1) - 4(x - 1)^2 + 2(x - 1)^3$.

Teacher answers & working

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EA2

Find each Taylor expansion about the stated point.

Find the Taylor expansion of ln x about x = 1 up to the cubic term.

Answer:

$(x - 1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}$

Working:

Let $h = x - 1$, so $x = 1 + h$.

Rewrite the function: ln x = ln(1 + h).

Use $\ln(1+h) = h - \frac{h^2}{2} + \frac{h^3}{3} - \dots$.

Substitute h = x - 1: $\ln x = (x - 1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \dots$.

Answer: $(x - 1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3}$.

EB1

Use a Taylor expansion about the stated point to approximate the value.

Use a Taylor expansion to approximate 1 2.1 .

Answer:

0.476187

Working:

Use the Taylor expansion for 1/x about x = 2.

Let h be the distance from the expansion point. Here h = 0.1.

Use $\frac{1}{2} - \frac{h}{4} + \frac{h^2}{8} - \frac{h^3}{16}$.

Substitute h = 0.1.

$\frac{1}{2.1} \approx \frac{1}{2} - \frac{0.1}{4} + \frac{(0.1)^2}{8} - \frac{(0.1)^3}{16}$.

Calculate this expression to get $\frac{1}{2.1} \approx 0.476187$.